

How Coding Agents *Fail* Their Users

A Large-Scale Analysis of Developer-Agent Misalignment in 20,574 Real-World Sessions

Ningzhi Tang¹, Chaoran Chen¹, Gelei Xu¹, Yiyu Shi¹, Yu Huang², Collin McMillan¹, Tao Dong³, Toby Jia-Jun Li¹

¹University of Notre Dame · ²Vanderbilt University · ³Google · ntang@nd.edu, toby.j.li@nd.edu

01 · MOTIVATION

Why real sessions?

Existing studies of coding-agent failures replay benchmark trajectories. We study misalignment as developers experience it in real-world sessions.

DEFINITION

Misalignment is a breakdown in developer-agent collaboration made visible by the developer's correction or pushback.

INSTRUCTION what developers explicitly ask for

INTENTION what they actually want

02 · DATA

20,574 real sessions, two sources

BUILT ON OUR PRIOR WORK · ASE 2026

Programming by Chat

Characterized developers' prompt behavioral intents, session structures, and interaction dynamics.



SPECTORY

14,789

1,441 repos

IDE + CLI · Sep 2024–Apr 2026

An IDE extension writes session as Markdown to .specstory/history/ and commits it to the repo.

ENTIRE.IO

5,785

198 repos

CLI · Jan–Apr 2026 · via SWE-chat

Git hooks save sessions as JSONL to branch entire/checkpoints/v1 and sync them on push.

20,574 SESSIONS · 1,639 REPOS · BY AGENT

IDE	12,231	CLI	8,343
Cursor	3,234	Claude Code	6,648
GitHub Copilot	366	OpenCode	624
Unknown *	8,631	Codex	517
		Gemini CLI	39
		Cursor CLI	32
		Unknown *	483

* agent identity missing from early SpecStory exports

03 · METHOD

A three-stage LLM pipeline

ALL STAGES · GPT-5.4

① EXTRACT

29,896

candidate episodes

The LLM scans each full session and grounds every episode in specific log turns.

EACH RECORD

name	goal	description	evidence
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② VALIDATE

16,118

kept (53.9%)

A second pass drops unsupported claims.

0.93 precision **1.77/2** mean recall rating

42.3% Normative prior bias

flags behavior the developer never objected to

57.7% Observational blind spots

relies on context not in the log

③ ANNOTATE

4 axes

per episode

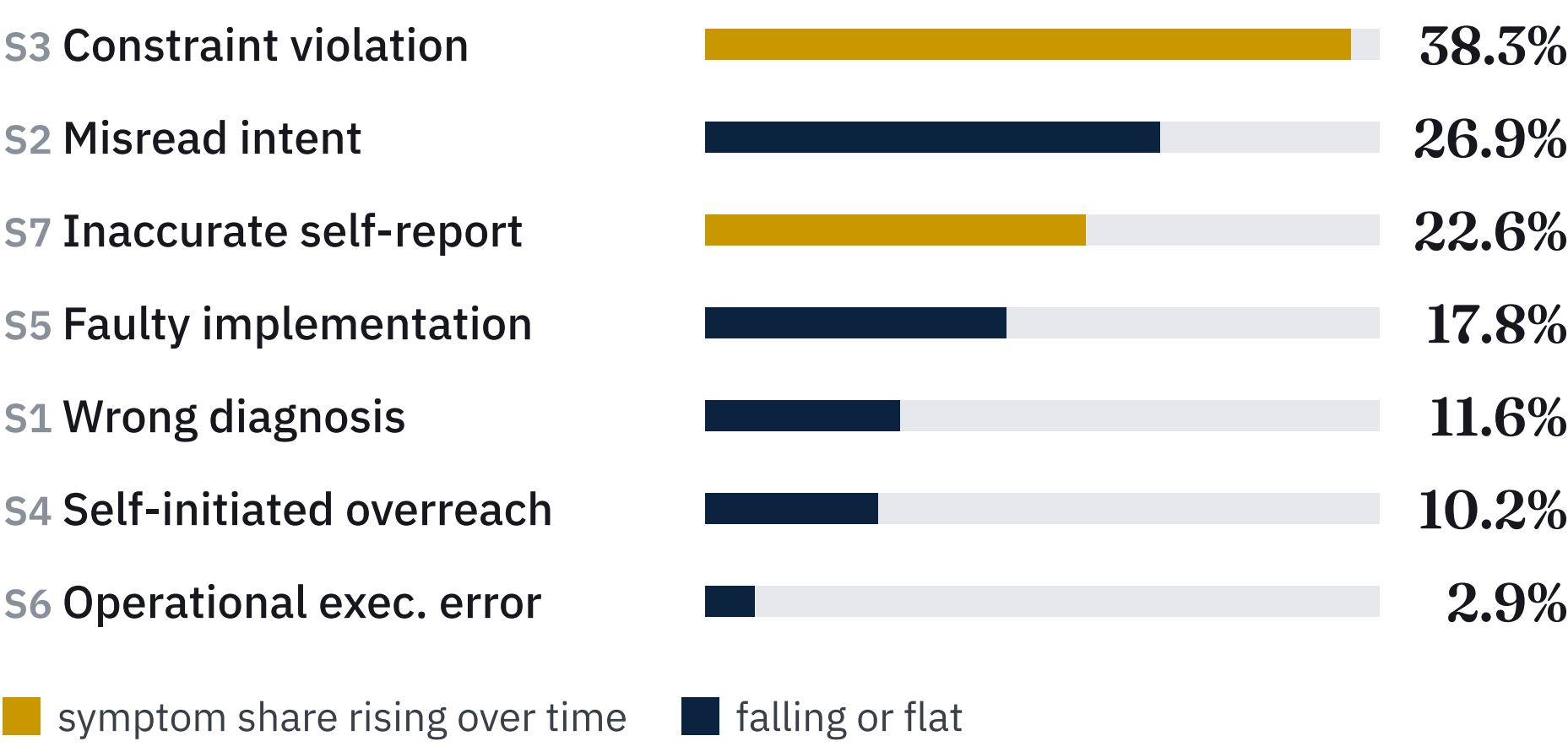
Symptom · cause · outcome · resolution.

0.78 human agreement **0.82** judge accuracy

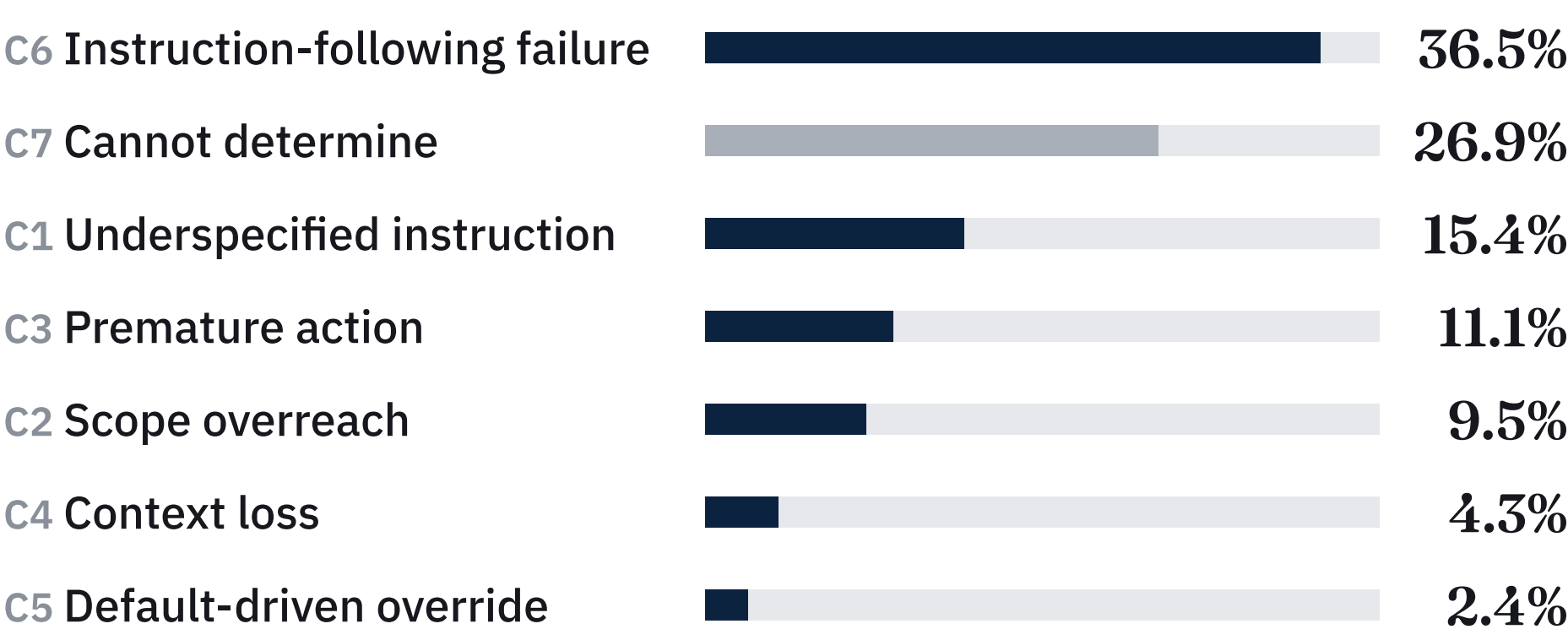
RQ1 · FORMS & CAUSES

What goes wrong, and why

SYMPTOMS · WHAT GOES WRONG



CAUSES · WHY

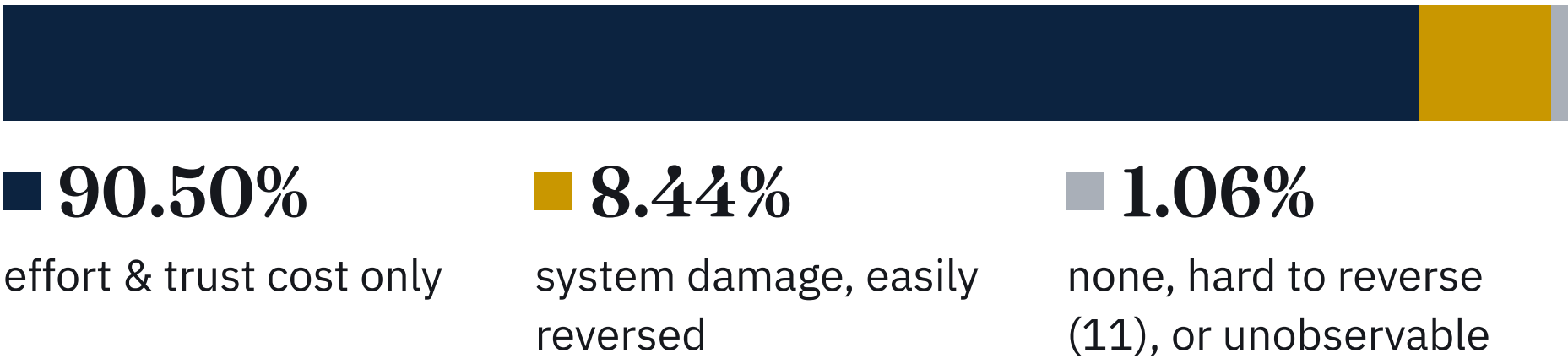


RQ2 · OUTCOMES & RESOLUTION

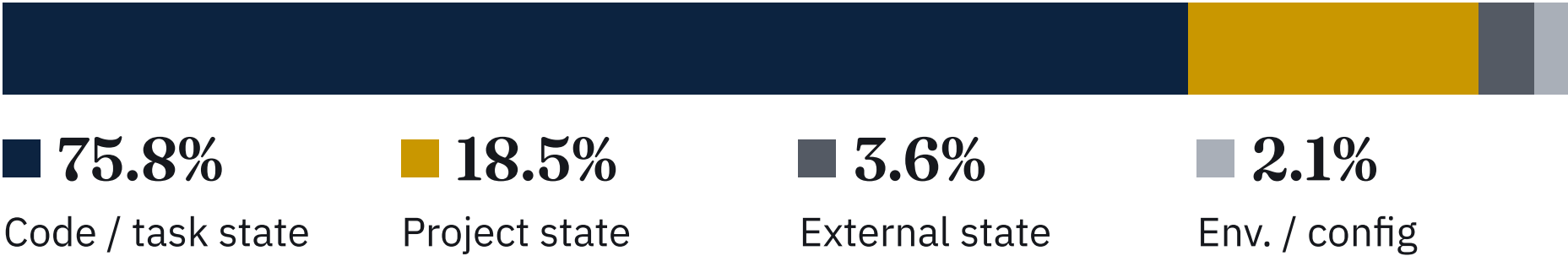
Cheap damage, costly attention

Most episodes cause no lasting system damage. The cost falls on the developer's time and trust.

DAMAGE SEVERITY · ALL 16,118 EPISODES



WHERE DAMAGE LANDS · 1,372 EPISODES



WHO FIXES IT? · 1,504 EPISODES WITH A VISIBLE FIX



EXAMPLE

One episode, from log to labels

EXTRACTED RECORD

name Changed Slide Orientation on an Unverified Assumption

goal intention

description The developer asked why slide 2 was landscape; the agent changed it to 4:3 portrait instead of explaining.

evidence T7 user "why is slide 2 showing landscape"
T8 agent "Fixed by explicitly setting the aspect ratio to 43 (4:3 portrait)..."
T9 user "i want the whole presentation 16:9 landscape"

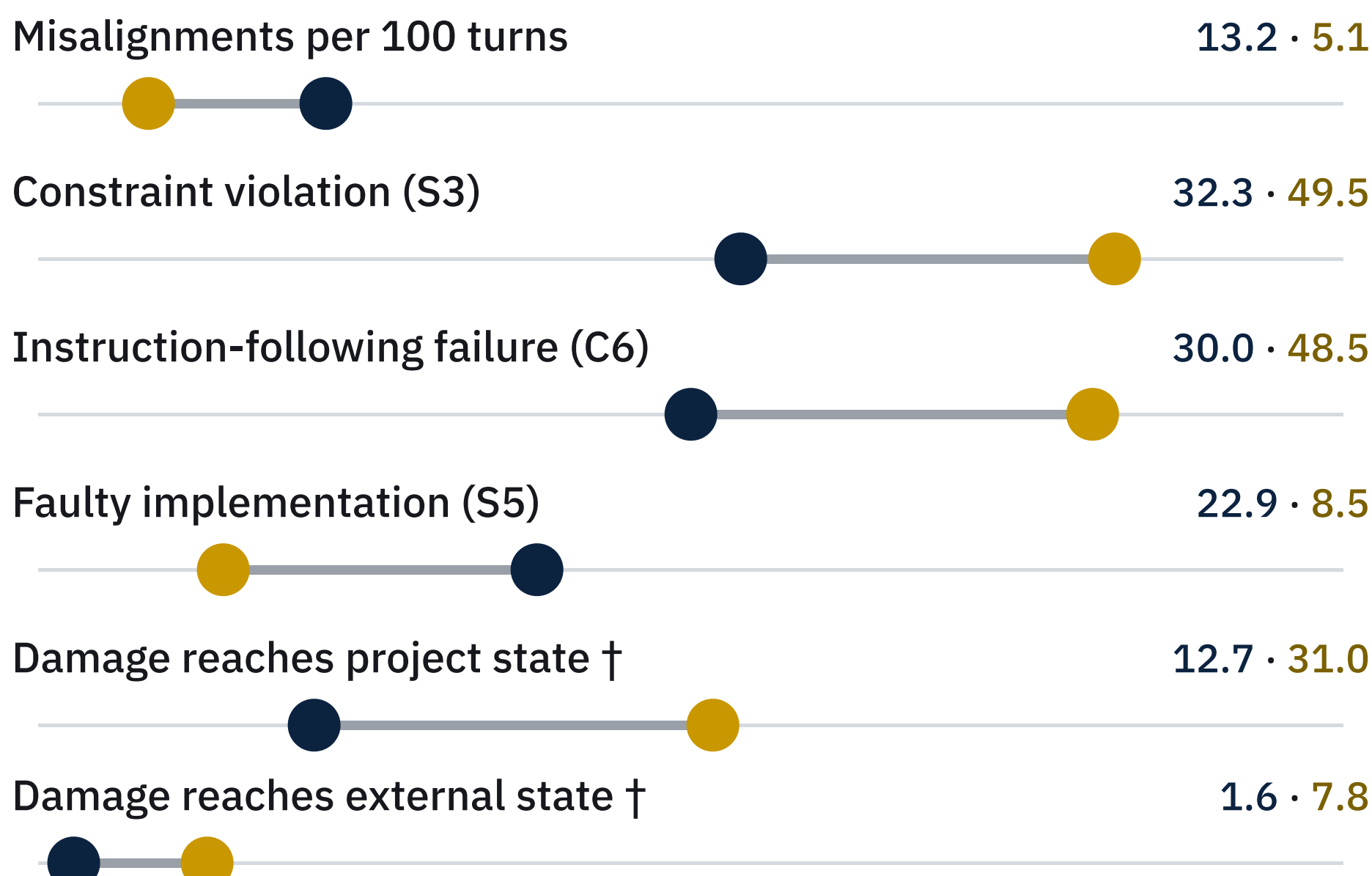
S4 OVERREACH S2 MISREAD INTENT C2 SCOPE OVERREACH

RQ3 · IDE VS. CLI

CLI agents break rules; IDE agents break code

Interaction shapes attention: IDE developers watch the code; CLI developers hand over control and watch the rules.

● IDE · median 3 turns/session ● CLI · median 5 turns/session



RQ4 · STRUCTURE & TIME

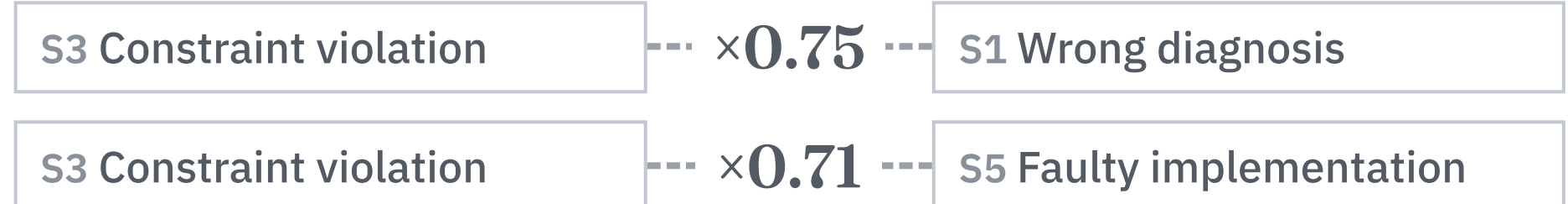
Misalignment co-occurs, persists, and shifts

WITHIN A SESSION · CO-OCCURRENCE LIFT

Above chance



Below chance



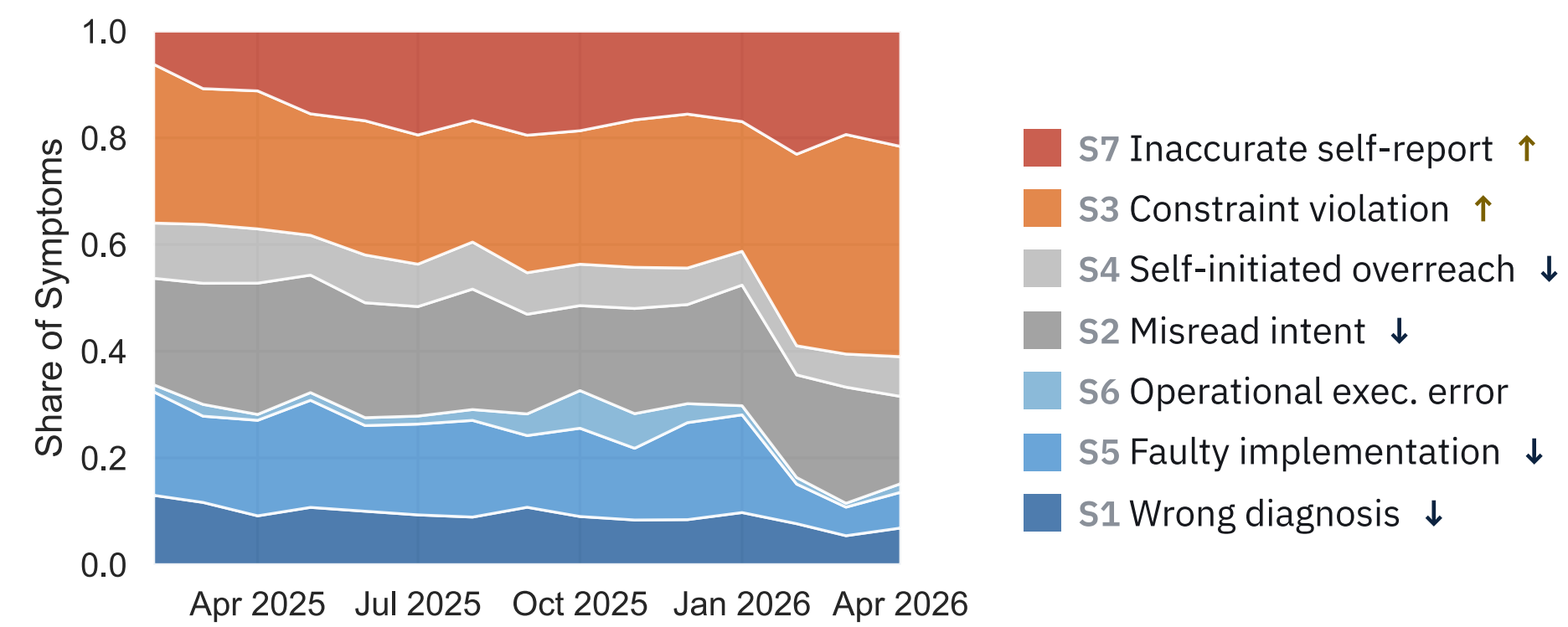
lift = observed ÷ expected co-occurrence; ×1 = chance

ACROSS SESSIONS · SAME REPO

+54% higher probability of misalignment in the next session (0.519 after a misaligned session vs. 0.336 otherwise)

OVER TIME · FEB 2025 → APR 2026

Misalignment per user turn declines ≈ 0.008 per month.



Explore the data

INTERACTIVE DASHBOARD
coding-agent-misalignment.netlify.app

CODE, LABELS & NOTEBOOKS
github.com/ND-SaNDwichLAB/coding-agent-misalignment

TAKEAWAY 1

Interaction-level failures are the new gap

Code-level symptoms decline in share, while constraint violations and inaccurate self-reporting rise.

TAKEAWAY 2

Today's safety rests on the developer

Most visible fixes require real-time developer pushback. That oversight may not scale to longer-running background agents.

TAKEAWAY 3

Session logs are a behavioral signal

The pipeline could turn live-session misalignment into evaluation cases and training signals for agent teams.